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Multi-Criteria Critical Path Analytics for Operational Risk Management in Multi-Tier Supply Chains

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Abstract: Multi-tier supply chains are exposed to operational disruptions that can spread through interconnected supplier–customer relationships. Existing risk assessments often rank individual firms or links without considering whether these relationships form continuous routes of concentrated vulnerability. This study investigates critical-path identification as a decision-analytics problem in which relationship-level performance and network structure are considered jointly. A two-phase framework was developed by integrating the Criteria Importance Through Intercriteria Correlation (CRITIC) method with evolutionary path optimization. In the first phase, CRITIC was used to derive objective weights for 14 operational performance criteria and to calculate a criticality index for each supplier–customer link. In the second phase, two optimization models were formulated to identify the path with the highest cumulative criticality and the path with the highest average link criticality subject to a minimum path length. The framework was applied to an automotive supply chain consisting of 19 enterprises and 35 directed links. The cumulative model identified a five-link path with a total criticality of 3.274 and an average criticality of 0.655, whereas the average-criticality model identified a four-link path with a total criticality of 2.738 and an average of 0.684. Both models selected the same initial relationship but produced different subsequent routes. The results indicate that link-level rankings alone cannot identify the most critical continuous route because path selection also depends on connectivity, topological position, and the optimization objective. The framework provides a reproducible basis for prioritizing supplier relationships, directing monitoring resources, and selecting risk-mitigation measures across multi-tier supply networks.

Keywords: Multi-tier supply chain; Critical path analytics; Operational risk; Criteria Importance Through Intercriteria Correlation method; Evolutionary optimization; Decision support

1 Introduction

A supply chain represents a complex system consisting of enterprises that have common goals, such as establishing effective cooperation, maintaining a competitive position in the market, ensuring business security and, last but not least, achieving the desired profit. Enterprises connected by strong partnership relationships can strengthen resilience in today's uncertain business environment, as well as jointly address potential challenges imposed by modern business [1, 2]. The process of coordinating and managing relationships between enterprises in the supply chain is referred to as supply chain management (SCM).

SCM is based on the joint efforts of management teams of all enterprises in the chain with the aim of optimizing the flow of materials and information through the supply chain [3]. This concept has developed to such an extent that it has become a complex and continuously developing scientific discipline, which continues to develop and improve. One of the key tasks of SCM is to coordinate the business processes of enterprises, i.e., members of the chain, with the aim of reducing waiting times, unnecessary costs, inventories, and other activities that are non-value-adding to enterprises. The primary motivation for the emergence and development of this concept is the effort to meet the requirements of customers (consumers, clients, or users) [4, 5].

A particularly important aspect of SCM relates to the operational risks to which individual participants, as well as the entire supply chain, are exposed [6]. Operational complexity across modern supply chains, the large number of participants, inter-enterprise dependencies, and escalating business, technological, and market requirements create conditions in which a disruption at a single participant can propagate toward other entities in the chain. Operational risks may originate from shifting customer requirements, technological advancements, economic conditions, market competition, quality variations, supplier reliability, resource availability, and other factors affecting business stability. Consequently, the identification and assessment of critical links and paths within the supply chain represent fundamental tasks in modern supply chain research.

Within this context, particular attention should be given to supplier–customer relationships between enterprises, rather than focusing exclusively on isolated entities. Each supplier–customer relationship captures a specific physical material flow and represents a potential channel through which disruptions can propagate to other supply chain segments. However, the criticality of an individual link does not necessarily mirror the criticality of the complete path through the supply chain. A link with the highest individual criticality may form part of a route whose remaining edges possess low criticality scores. Conversely, several interconnected links characterized by moderate or high criticality may form a continuous path representing a substantially greater source of overall risk exposure. Therefore, alongside individual link evaluation, the structural topology of the supply chain and interconnections among its participants must be considered simultaneously.

Identification of a critical path across a supply chain constitutes a complex combinatorial optimization problem due to graph topology and the existence of multiple feasible routes between participants. An additional challenge arises because criteria selection and weight determination directly govern link criticality assessments. Subjective criteria weighting causes evaluation outcomes to depend on individual decision-maker opinions, whereas neglecting inter-criteria correlations leads to information loss within the decision matrix. Furthermore, critical path identification requires solving an optimization model in which individual link criticality metrics and network structural constraints are integrated simultaneously.

The primary aim of this study is to develop an integrated methodology for identifying critical paths in supply chains, wherein link criticality is first determined objectively and subsequently used as input for network optimization. The proposed framework combines the Criteria Importance Through Intercriteria Correlation (CRITIC) [7–9] method for objective criteria weighting with the Evolutionary Solver for critical path identification. Through this design, multi-criteria assessment of supplier-customer relationships is directly linked to path optimization within the supply chain.

A specific objective of this research is to analyze critical paths from two complementary optimization perspectives. In the first model, total cumulative criticality across all links belonging to the selected path is maximized. In the second model, average link criticality along the path is maximized, thereby identifying routes where operational risk is concentrated across constituent links. By comparing these two model variants, common highly critical segments as well as structural divergence resulting from alternative path criticality definitions can be identified.

The main contribution of this study lies in connecting the objective assessment of supplier–customer relationship criticality with an evolutionary optimization approach for critical path identification in complex supply chains. Unlike traditional methodologies focused solely on ranking individual participants or links, the proposed framework explicitly incorporates their topological position and connectivity within the supply chain. By combining an analysis of cumulative and average criticality, the methodology presented in this article extends the field of supply chain analysis and allows for a more comprehensive interpretation of supply chain vulnerabilities.

Demonstrated through a multi-tier case study from the automotive sector, with interconnected participants and physical flows of materials, this methodology allows practitioners to detect the supply chain segments that require most attention for risk management and decision-making.

2 Literature Review

Within SCM, particular attention has been paid by researchers to the problem of risk management in supply chains. One of the studies that shows the significance and diversity of this problem is presented [10]. The authors of this study conducted a state-of-the-art analysis in this domain, highlighting that new risks in supply chains have emerged in recent years, caused primarily by the pandemic situation, but also by an unstable political and economic environment.

Supply chains are no longer conceptualized as mere collections of independent firms, but are increasingly modeled as interconnected graphs. For example, global automotive supply networks were analyzed by Yue and Zhong [11] through a multilayer network model integrated with centrality metrics and dynamic disruption tracking. The study showed that the spread of cascade effects from upstream levels through intermediate suppliers affects the further flow of the network.

A network-oriented approach was applied in the study by Liu and Ren [12]. Objective link weighting was combined with topological analysis. Primarily, composite edge weights were determined by applying the CRITIC method through the evaluation of the considered criteria, such as transportation distance, transit time, and geopolitical exposure. Next, an enhanced weighted k-shell model was deployed to detect critical nodes. Integration of objective link weights into network-level criticality modeling was successfully demonstrated. Nevertheless, the analysis was focused on isolated critical nodes rather than full path identification across the supply chain.

Automated identification of critical paths represents a distinct research direction in supply chain research. An automated framework was introduced by Han et al. [13], wherein supply chain knowledge graphs were encoded and processed using large language models (LLMs). Through that approach, critical supply chain paths were successfully identified alongside natural language reasoning. However, this graph-reasoning approach differs fundamentally from mathematical optimization, as path identification relies on language model-based heuristics rather than on explicitly formulated optimization algorithms.

In complex supply chain environments, criticality evaluation often requires the integration of heterogeneous performance metrics. Consequently, Multi-Criteria Decision-Making (MCDM) methodologies are broadly applied across supplier evaluation and risk modeling. Among objective weighting frameworks, the CRITIC method formulated by Diakoulaki et al. [7] is widely used. Criteria weights are derived by evaluating both criteria variance and inter-criteria correlations. Unlike subjective weighting techniques, criteria importance is determined directly from the underlying empirical decision matrix.

Practical applications of the CRITIC method across supply chain domains have been reported in multiple studies. Thermal coal suppliers were evaluated in the study by Zhong et al. [14]. The authors addressed limitations of the CRITIC method, such as conflicting indicator ranges, through a specially defined procedure. Also, the limitations of the standard CRITIC method were highlighted by Zhang et al. [15], particularly regarding the limitations of Pearson's correlation coefficient in addressing non-linear dependencies in the data. The authors of this study introduced an additional procedure for testing dependencies in order to overcome this problem.

The application of metaheuristic approaches in solving supply chain optimization problems is widely represented in the literature. The authors have used numerous methods to address a wide range of highly diverse problems. A risk-oriented supply chain optimization framework was presented by Nezamoddini et al. [16], where a genetic algorithm [17–19] was combined with an artificial neural network. Decisions across strategic, tactical, and operational horizons under supply chain risk were considered, while optimal solutions were located through evolutionary search mechanisms. The applicability of evolutionary search to complex combinatorial problems was thus demonstrated.

Combinatorial problems within constrained supply chain networks are particularly suitable for evolutionary search techniques. Genetic operators have received extensive attention in network design, production-distribution scheduling, vehicle routing, and facility allocation. Nevertheless, traditional optimization models are guided predominantly by operational or financial objectives, such as cost minimization, inventory targets, capacity limits, or profit maximization. Explicit path identification based on multi-criteria supplier–customer link criticality remains largely unaddressed in the existing literature.

Two well-established yet distinct research streams can be identified in the literature. In the first stream, supply chain risk and criticality are evaluated using MCDM techniques, objective weighting, network centrality, Bayesian networks, or risk propagation modeling. In the second stream, network configuration and operational decisions are optimized using evolutionary algorithms. Research explicitly linking objective multi-criteria assessment of individual supplier–customer links with path optimization through the supply chain remains scarce.

This distinction is important. Individual supplier–customer relationship criticality does not necessarily determine complete path criticality toward the focal enterprise. A single high-risk link may exist within a route of low overall vulnerability. Conversely, multiple moderately critical links can accumulate into a more vulnerable continuous path. Evaluating links in isolation is therefore insufficient for identifying the most critical route across a multi-tier supply chain.

Automotive supply chains provide a suitable empirical setting due to their multi-tier architecture and inter-enterprise dependencies. Risks in the Turkish automotive sector were investigated by Çıkmak and Ungan [20] via expert interviews and Bayesian network modeling. Inter-risk dependencies and mitigation impacts were highlighted. In the study by Blos et al. [21], a case study was conducted in the field of risk management in the automotive and electronic industries in Brazil. Although the study has practical significance in this domain, the research problem was not optimization-oriented.

In the study by Fraser et al. [22], the transparency of a multi-tier supply chain was analyzed. The study showed that there is insufficient transparency among lower-tier suppliers, which significantly reduces the control and influence of the focal organization. This conclusion indicates the need to develop models that can evaluate and optimize supply chains with complex structures.

Prior studies have separately addressed risk assessment, objective criteria weighting, critical node identification, risk propagation, and evolutionary optimization. However, integrated frameworks remain lacking wherein: 1) Individual supplier–customer link criticality is objectively evaluated across multiple performance criteria; 2) Link criticality indices are embedded into a mathematical path optimization formulation; 3) Alternative path criticality formulations are explicitly compared.

The issue of resilience in supply chain network structures and the occurrence of disruptions has already been analyzed in the relevant literature [23–25]. Recent studies investigating multi-tier network structures indicate the existence of critical nodes that may subsequently affect the entire network and supply vulnerability [26–28]. Furthermore, the literature also includes optimization-oriented studies in this domain aimed at supporting decision-making regarding supply chain recovery [29, 30].

This gap is addressed in the present study through an integrated approach combining the CRITIC method with evolutionary optimization. Objective criteria weights and link criticality indices are first derived via CRITIC for each supplier–customer relationship. These indices serve as parameters for two complementary optimization models. Maximum cumulative criticality is targeted by the first model, whereas maximum average link criticality subject to a minimum length constraint is identified by the second. Structural positioning and connectivity are thus explicitly incorporated into path evaluation. Furthermore, structural insights into whether risk is concentrated within isolated links or distributed along an extended path are revealed through comparative analysis.

3 Methodology

A methodology for identifying critical paths in a supply chain was developed in this study. The methodology is based on a two-phase approach that combines an objective assessment of the criticality of individual supplier–customer links and optimization using an evolutionary approach. In the first phase, the criteria weights are determined using the CRITIC method, based on which the criticality of the link between a supplier and a customer is subsequently calculated, while in the second phase, critical paths are identified using the evolutionary algorithm available in the Solver package (Microsoft Excel). An overview of the proposed methodology is presented in Figure 1.

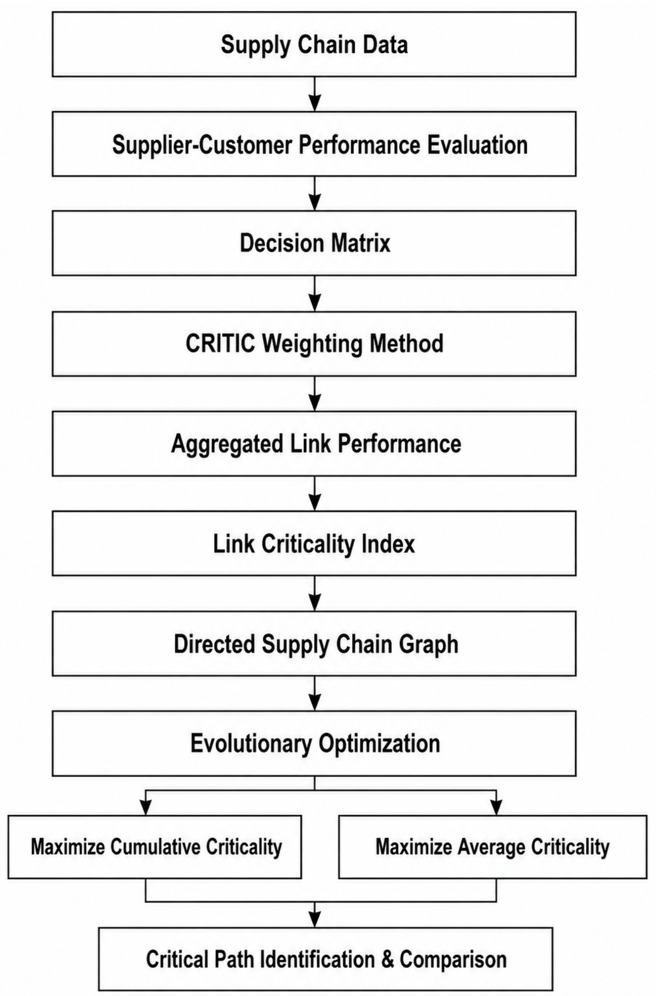

Figure 1. Overview of the proposed methodology for critical path identification

The proposed methodology is based on modeling the supply chain as a directed graph. Supply chain participants are represented as nodes, while their connections, i.e., material flows, are represented as edges (see Figure 2). The performance of the link between two companies in the supply chain (supplier–customer) is determined based on a set of defined criteria describing the performance of this link. Based on the data collected through the questionnaire, a decision matrix is formed and used as the basis for determining the criticality of each individual link.

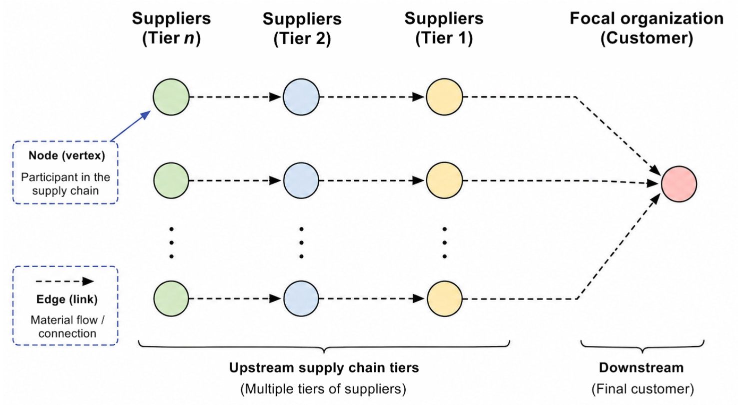

Figure 2. Supply chain representation as a directed graph

As previously mentioned, the CRITIC method was used in the first phase of the proposed methodology. This method is based on the information content of the criteria, while the performance of the links between suppliers and customers is assessed by the customers and considered using 14 indicators, i.e., criteria. The information value of each criterion is determined based on the intercorrelation between the criteria and the dispersion of values within each criterion. After determining the criteria weights, a criticality index is calculated for each link in the supply chain. These values represent the input parameters for the second phase of the proposed methodology.

The problem is stated as a combinatorial optimization problem with constraints that ensure the formation of a continuous, non-branching path between the starting and ending nodes. The Evolutionary Solver, which is an integral part of the Microsoft Excel software package, was used to solve this optimization model. It is important to emphasize that this algorithm belongs to the class of evolutionary algorithms and is based on the principles of genetic algorithms. The algorithm is based on searching the space of possible solutions through repeated evolutionary operations, i.e., selection, crossover, and mutation.

Two complementary optimization models are considered in this study. The first model determines the path with the highest total criticality; in other words, it seeks the maximum sum of the criticality values of all links belonging to a particular path. The aim is to identify the path with the highest cumulative criticality in the supply chain. On the other hand, the second model maximizes the average criticality of the links, thereby reducing the advantage of longer paths that may arise when maximizing total criticality. However, to avoid favoring trivially short paths with a small number of highly critical links, a constraint on the minimum number of links belonging to the path is introduced.

The aim of the proposed approach is to analyze the same supply chain from two perspectives. By comparing the obtained results, it is possible to identify common critical segments of the chain, as well as potential differences between the identified paths. In this way, decision-makers obtain a broader view of the supply chain and the criticality of its operations.

3.1 Problem Description and Supply Chain Network Representation

The supply chain structure is represented as a directed graph $G = (V, E)$. In this network formulation, supply chain participants are denoted by a set of nodes v, $v = \{1, \ldots, V\}$, whereas directed links between suppliers and customers are denoted by a set of edges e, $e = \{1, \ldots, E\}$. Each directed edge $e = (i, j) \in E$ represents the material flow moving downstream from supplier i to customer j.

For each link e, operational performance is assessed with respect to a defined set of criteria k, $k = \{1, \ldots, K\}$. Based on collected performance evaluations, the initial decision matrix is defined as $X = [x_{ek}]_{m \times q}$, where x_{ek} represents the performance score of link e evaluated against criterion k. In the considered optimization problem, all criteria are defined as benefit-type indicators; hence, a higher value reflects superior link performance.

The primary objective is defined as the identification of a continuous, unbroken path $P \subseteq E$ extending from an algorithmically determined starting node to the focal organization. This path is selected such that total criticality exposure is maximized according to the specified objective function.

3.2 Criticality Assessment Using the Criteria Importance Through Method

To reduce subjectivity in criteria weighting, the CRITIC method is employed to determine the weights objectively from the available assessment data. First, min-max normalization is applied to the initial decision matrix X. Since all evaluated criteria are of the benefit type, normalized values r_{ek} are calculated via Eq. (1):

$$r_{ek} = \frac{x_{ek} - \min_e x_{ek}}{\max_e x_{ek} - \min_e x_{ek}} \tag{1}$$

For each criterion k, the standard deviation σ_k and the Pearson correlation coefficient ρ_{kl} between criteria k and l are calculated. The overall information content C_k contained within criterion k is determined as:

$$C_k = \sigma_k \sum_{k=1}^{K} (1 - \rho_{kl}) \tag{2}$$

The relative weight ω_k of criterion k is subsequently calculated by normalizing information values across all criteria:

$$\omega_k = \frac{C_k}{\sum_{k=1}^{q} C_k} \tag{3}$$

where, $\sum_{k=1}^{q} \omega_k = 1$.

Based on the derived criteria weights, the aggregated performance score P_e for each link is calculated as:

$$P_e = \sum_{k=1}^{K} \omega_k \cdot r_{ek} \tag{4}$$

Since a higher P_e value signifies better overall link performance, supply chain vulnerability is modeled by inverting this metric. Accordingly, the link criticality index c_e is defined as:

$$c_e = 1 - P_e \tag{5}$$

Consequently, higher c_e values indicate greater criticality along link e. The obtained c_e values are directly utilized as input parameters for the secondary optimization phase.

3.3 Mathematical Model for Critical Path Identification

For each link $e \in E$, a binary decision variable y_e is defined to indicate path inclusion:

$$y_e = \begin{cases} 1, & \text{if link } e \text{ belongs to the selected path} \\ 0. & \text{otherwise} \end{cases} \tag{6}$$

For every node $i \in V$, incoming and outgoing active links are defined as IN_i and OUT_i, respectively:

$$IN_i = \sum_{e \in \delta^-(i)} y_e, \quad OUT_i = \sum_{e \in \delta^+(i)} y_e \tag{7}$$

where, incoming and outgoing edge sets associated with node i are designated by $\delta^-(i)$ and $\delta^+(i)$, respectively.

To prevent structural path branching and ensure a simple linear route, the following upper-bound constraints are imposed:

$$IN_i \leq 1, \quad \forall i \in V \tag{8}$$

$$OUT_i \leq 1, \quad \forall i \in V \tag{9}$$

For each intermediate node along the selected path, the flow conservation condition must be satisfied:

$$IN_i = OUT_i \tag{10}$$

At the selected starting node s and terminal node t, net outflow and net inflow conditions are enforced:

$$OUT_s - IN_s = 1 \tag{11}$$

$$IN_t - OUT_t = 1 \tag{12}$$

Additionally, path length is bounded from below to avoid short, trivial routes:

$$\sum_{e \in E} y_e \geq L_{\min} \tag{13}$$

where, $L_{\min}$ denotes the minimum allowable number of links in the selected path. In this formulation, $L_{\min}$ = 4 is specified.

3.4 Evolutionary Optimization Model

The formulated combinatorial optimization problem is solved using the Evolutionary Solver integrated into the Microsoft Excel software package. This solver belongs to the class of evolutionary algorithms and is based on genetic algorithm principles, including population-based search mechanisms, selection, crossover, and mutation operators.

The decision variables are represented by binary vectors, while the feasible solution space is bounded by constraints Eq. (8)–Eq. (12). Due to the stochastic nature of evolutionary operators, each optimization model was executed in five independent runs. The same optimal solution was obtained in all five runs and was therefore selected as the final solution.

The evolutionary optimization procedure was conducted using the Evolutionary Solver integrated into Microsoft Excel. Specific algorithmic parameters were configured to control the search process. A population size of 500 was used. A mutation rate of 0.20 was applied alongside a convergence threshold of 0.0001. A maximum runtime without improvement was set to 120 s. Different random seeds were used across the five runs to maintain stochastic variation, while the optimization settings were kept unchanged across runs. Since the decision variables are binary, integer constraints were explicitly imposed on all link-selection variables.

3.5 Optimization Models

Based on the calculated criticality indices c_e, two complementary optimization models are formulated. Both models utilize identical binary decision variables and constraint sets Eq. (8)–Eq. (12), while differing in their objective function formulations.

3.5.1 Maximum cumulative criticality model

In the first model, the path possessing the maximum cumulative criticality is identified:

$$\max Z_1 = \sum_{e \in E} c_e y_e \tag{14}$$

The objective function in Eq. (14) favors the path along which the highest overall level of criticality is accumulated.

3.5.2 Maximum average criticality model

The second model identifies the path with the maximum average criticality:

$$\max Z_2 = \frac{\sum_{e \in E} c_e y_e}{\sum_{e \in E} y_e} \tag{15}$$

Unlike the first model, the objective function in Eq. (15) normalizes cumulative criticality by the number of selected links, thereby eliminating the structural bias toward longer paths. Simultaneously, the selection of trivially short paths consisting of only a small number of highly critical links is restricted by constraint $L_{\min}$.

4 Case Study and Results

4.1 Case Study Description

Empirical validation of the proposed methodology was conducted through a multi-tier supply chain case study within the automotive sector. The focal analytical setting was centered on the production system of an automotive seat assembly. Within this supply chain, various raw materials, structural subassemblies, and functional components supplied across multiple upstream tiers are integrated by a single focal enterprise. High manufacturing complexity guided the selection of this specific assembly. Its production requires the simultaneous integration of diverse physical inputs, such as structural steel frames, molded polyurethane foam parts, upholstery fabrics, mechanical adjustment mechanisms, and various supporting hardware components.

The analyzed supply chain comprises 19 interconnected enterprises acting as supply chain participants. For the case study, these enterprise nodes are denoted as, whereas the 35 inter-organizational supplier-customer relationships are designated as. Accordingly, graph nodes (defined in Section 3.1) correspond directly to enterprises, while graph links correspond to relationships. Each link is defined by its specific supplier–customer relationship and physical material flow between enterprises. For example, material flow from supplier enterprise to customer enterprise is designated by.

Positioned at the downstream terminal of the analyzed supply chain, focal enterprise functions as the focal integrator. Upstream supplier enterprises located across different tiers are represented by. Inter-firm relationships among participants are captured by 35 directed links. The supplier-customer relationships are presented in Table 1. In the considered network, the focal organization is designated as the terminal node for any identified path. On the other hand, the starting node is not specified in advance but is determined by the algorithm through the analysis of the network.

Table 1. Supplier–customer relationships in the analyzed supply chain

Link	Supplier	Customer	Link	Supplier	Customer	Link	Supplier	Customer
L_1	S_2	S_1	L_{13}	S_{11}	S_5	L_{25}	S_{18}	S_{11}
L_2	S_3	S_1	L_{14}	S_{11}	S_6	L_{26}	S_{18}	S_{12}
L_3	S_4	S_1	L_{15}	S_{12}	S_6	L_{27}	S_{19}	S_{12}
L_4	S_5	S_1	L_{16}	S_{13}	S_7	L_{28}	S_{19}	S_{13}
L_5	S_6	S_1	L_{17}	S_{14}	S_7	L_{29}	S_{18}	S_{13}
L_6	S_7	S_2	L_{18}	S_{14}	S_8	L_{30}	S_{18}	S_{14}
L_7	S_8	S_2	L_{19}	S_{15}	S_8	L_{31}	S_{19}	S_{15}
L_8	S_8	S_3	L_{20}	S_{15}	S_9	L_{32}	S_{19}	S_{16}
L_9	S_9	S_3	L_{21}	S_{16}	S_9	L_{33}	S_{18}	S_{16}
L_{10}	S_9	S_4	L_{22}	S_{16}	S_{10}	L_{34}	S_{19}	S_{17}
L_{11}	S_{10}	S_4	L_{23}	S_{17}	S_{10}	L_{35}	S_{19}	S_{18}
L_{12}	S_{10}	S_5	L_{24}	S_{17}	S_{11}	-	-	-

Note: "L" denotes a link, "S" denotes a supplier, and "-" indicates that no corresponding link is available.

A non-linear structure is exhibited by the analyzed supply chain. Individual upstream supplier enterprises frequently deliver materials to multiple downstream participants. Conversely, single customer enterprises routinely receive inputs from several distinct suppliers. For instance, direct components are delivered to focal enterprise by five immediate tier-1 supplier enterprises. Further upstream, tier connections branch into alternative supply routes. Consequently, numerous continuous, feasible paths extending from upstream supplier enterprises toward the focal enterprise are formed within the supply chain.

All 35 supplier–customer links representing inter-organizational relationships were evaluated against 14 performance criteria. To reflect operational realities, evaluations were conducted strictly from the customer perspective. Each direct supplier enterprise was evaluated by its immediate purchasing enterprise. With this approach, each supplier-customer relationship was scored by a decision-maker directly involved in managing that specific commercial interface.

For each enterprise acting as a customer within the supply chain, performance assessments were provided by a single responsible manager having direct operational knowledge of supplier performance and inter-organizational relationships. Depending on the internal organizational structure of the customer enterprise, positions such as procurement manager, supply chain director, production manager, quality manager, or logistics coordinator were included. Exactly one qualified manager per enterprise was surveyed to obtain a single managerial assessment for every supplier–customer relationship from the customer side.

All criteria were assessed based on the manager's professional experience, operational knowledge, and available information regarding the respective supplier and the established supplier–customer relationship.

Scoring was executed on a nine-point scale ranging from 1 to 9. Because all 14 criteria were formulated as benefit-type indicators, a score of 1 reflects the least favorable performance, whereas a score of 9 denotes the most favorable supplier performance for the observed supplier–customer relationship. Intermediate performance levels are represented by integer values between these bounds.

The adoption of the 14 performance criteria evaluated in this study was based on the supply chain performance framework proposed by Komatina and Živković [31]. Within that framework, overall supply chain performance is decomposed into three core dimensions: demand uncertainty, quality, and added value. Specific contextual adaptations were subsequently made to align these criteria with the operational characteristics of the analyzed automotive supply chain. The 14 performance criteria evaluated in the study comprise:

1. Customer uncertainty management: Refers to the supplier enterprise's ability to manage operational risks resulting from fluctuations in customer demand and purchasing behavior. The criterion considers the supplier's ability to accommodate demand variability, changes in product specifications, and unexpected changes in order volumes.
2. Technological uncertainty management: Refers to the supplier enterprise's ability to manage operational risks associated with technological changes, process modifications, and changing technical requirements. It also considers the supplier enterprise's ability to follow technological trends and adapt its production capacity when necessary.
3. Economic resilience: Refers to the supplier enterprise's ability to maintain stability and performance under changing external economic conditions. The criterion considers unit production costs, financial liquidity, and resilience to broader macroeconomic changes.
4. Market competition: Refers to the ability of the supplier enterprise to maintain a strong and stable competitive

position within its market segment. Price competitiveness, quality, delivery punctuality, production flexibility, and responsiveness to customer requirements are considered.

5. Integration of institutional norms: Refers to the extent to which regulatory, industry-specific, and formal operational standards are integrated into the supplier enterprise's daily operations. The criterion also considers compliance with legal requirements and quality standards relevant to the customer–supplier relationship.
6. Quality competitiveness: Refers to the ability of the supplier enterprise to maintain product and process quality that provides a competitive advantage in the market. Quality consistency, product reliability, and compliance with customer specifications are considered.
7. Critical success factors: Refers to the supplier enterprise's ability to manage the key operational factors required for successful cooperation with the customer enterprise. These include quality control, delivery reliability, operational flexibility, cost efficiency, communication, and responsiveness.
8. Strategic alignment: Refers to the degree of alignment between the long-term objectives of the supplier and customer enterprises. The criterion considers long-term relationship orientation, joint operational planning, and the supplier's willingness to support customer objectives.
9. Quality management paradigms and practices: Refers to the supplier enterprise's commitment to systematic quality management principles and practices. Process improvement, defect prevention, standardized quality inspection, and corrective action procedures are considered.
10. Stakeholder involvement: Refers to the level of cooperation and integration with key internal and external stakeholders involved in supplier operations. In addition to the customer–supplier relationship, the criterion considers involvement of workforce units, sub-tier suppliers, regulatory authorities, and industry associations.
11. Profit growth: Refers to the ability of the supplier enterprise to maintain continuous financial profitability. The criterion considers stable or increasing profit trends, long-term financial sustainability, operational cost efficiency, and the commercial viability of the customer–supplier relationship.
12. Asset utilization: Refers to the efficiency with which the supplier enterprise uses its physical assets, production capacity, and manufacturing equipment. Particular attention is given to the use of available plant capacities and technological assets to achieve defined operational targets.
13. Organizational credibility: Refers to the level of trust in the supplier enterprise's ability to fulfill its contractual commitments. Historical reliability, adherence to commercial agreements, operational transparency, and corporate reputation are considered.
14. Social responsibility: Refers to the extent to which ethical, social, and environmental responsibilities are integrated into the supplier enterprise's operations. The criterion covers ethical business practices, fair working conditions, community engagement, and the broader social impact of enterprise activities.

The collected managerial ratings were compiled into an initial decision matrix X of dimensions 35 × 14, providing the empirical baseline for the subsequent CRITIC-based link criticality assessment.

4.2 Criticality Assessment of Supply Chain Links

Objective weighting across the 14 performance criteria was derived by executing the CRITIC method according to the procedure defined in Section 3.2. Min-max normalization was first applied to the initial 35 × 14 decision matrix (Table 2). Subsequently, standard deviations for each criterion and pairwise Pearson correlation coefficients across criteria were computed.

Standard deviations ranging from 0.233 to 0.350 were recorded. Varying degrees of discriminatory information are thus reflected among the evaluated criteria. Based on the calculated information values and inter-criteria correlations, the following weight vector was derived:

$$\omega_k = (0.11, 0.10, 0.06, 0.05, 0.10, 0.05, 0.06, 0.09, 0.07, 0.09, 0.06, 0.05, 0.05, 0.06)$$

Composite performance scores P_e and criticality indices c_e for all 35 supplier-customer links were calculated using these derived weights. Criticality indices are bounded between 0 and 1. Values approaching 1 denote greater operational criticality within a given supplier–customer relationship between enterprises.

Substantial variations in criticality were observed across the analyzed supply chain links. The lowest criticality index was obtained for L_1 (c_1 = 0.065). Maximum criticality values of 1 were obtained for both L_{32} and L_{35}. Elevated criticality indices were also recorded for L_{33} (0.878), L_{28} (0.843), L_{27} (0.813), and L_{16} (0.805).

These isolated values provide valuable insights into individual supplier-customer relationship vulnerabilities. Nevertheless, simple link ranking based on standalone criticality is insufficient for identifying the most critical continuous path across the complete supply chain. A link with maximum individual criticality may belong to an overall route characterized by low risk across its remaining segments. Conversely, a continuous path formed by multiple moderately critical links can generate a substantially higher cumulative vulnerability. Consequently, the derived c_e metrics were deployed directly as input parameters within the two optimization models.

Table 2. The decision matrix

L	$k=$ 1	$k=$ 2	$k=$ 3	$k=$ 4	$k=$ 5	$k=$ 6	$k=$ 7	$k=$ 8	$k=$ 9	$k=$ 10	$k=$ 11	$k=$ 12	$k=$ 13	$k=$ 14
L_1	8	8	7	8	9	9	8	9	9	8	7	8	9	8
L_2	7	8	7	8	8	9	8	8	9	7	7	8	9	8
L_3	8	7	6	7	9	8	8	8	9	8	6	7	8	9
L_4	7	9	7	8	9	9	9	9	8	8	7	8	9	8
L_5	7	8	6	7	8	8	8	8	8	7	6	7	8	7
L_6	7	7	6	7	8	8	7	7	8	6	6	7	8	7
L_7	8	8	7	8	9	9	8	8	9	8	7	8	9	8
L_8	7	8	7	8	9	9	8	8	9	7	7	8	9	8
L_9	6	7	6	7	8	8	7	7	8	7	6	7	8	7
L_{10}	7	7	6	7	8	8	7	8	8	7	6	7	8	7
L_{11}	6	8	6	7	8	8	7	8	8	6	6	7	8	7
L_{12}	7	8	6	7	8	8	8	8	8	7	6	7	8	7
L_{13}	6	7	5	6	8	7	7	7	8	6	5	6	7	7
L_{14}	6	7	5	6	8	7	7	7	8	6	5	6	7	7
L_{15}	7	8	6	7	8	8	8	8	8	7	6	7	8	7
L_{16}	6	6	5	6	7	7	6	6	7	5	5	6	7	6
L_{17}	7	7	6	7	8	8	7	7	8	6	6	7	8	7
L_{18}	8	7	6	7	8	8	8	8	8	7	6	7	8	7
L_{19}	7	8	7	8	8	9	8	8	9	7	7	8	9	8
L_{20}	6	8	7	7	8	8	7	8	8	7	7	7	8	8
L_{21}	6	7	5	6	7	7	6	7	7	6	5	6	7	6
L_{22}	6	7	5	6	7	7	7	7	7	6	5	6	7	6
L_{23}	7	8	6	7	8	8	8	8	8	7	6	7	8	7
L_{24}	7	8	6	7	8	8	8	8	8	7	6	7	8	7
L_{25}	6	6	5	6	7	7	6	6	7	6	5	6	7	6
L_{26}	6	6	5	6	7	7	6	6	7	6	5	6	7	6
L_{27}	5	6	5	6	7	7	6	7	7	5	5	6	7	6
L_{28}	5	6	5	6	7	7	6	6	7	5	5	6	7	6
L_{29}	6	6	5	6	7	7	6	6	7	6	5	6	7	6
L_{30}	6	6	5	6	7	7	6	7	7	6	5	6	7	6
L_{31}	6	7	5	6	7	7	6	7	7	6	5	6	7	6
L_{32}	5	6	4	5	7	6	5	6	6	5	4	5	6	5
L_{33}	5	6	5	5	7	6	6	6	7	5	5	6	7	6
L_{34}	6	7	5	6	7	7	6	7	7	6	5	6	7	6
L_{35}	5	6	4	5	7	6	5	6	6	5	4	5	6	5

Note: "L" denotes a link, and k denotes the performance criterion index ($k = 1, \ldots, 14$).

4.3 Optimization Parameters

Execution of the optimization analysis was based on the supply chain structure defined by 19 enterprise nodes (S_1–S_{19}) and 35 directed links (L_1–L_{35}). A binary decision variable y_e was assigned to each link e. Path inclusion is indicated by y_e = 1, whereas non-selection is denoted by y_e = 0.

The primary numerical parameters for the optimization models were provided by the criticality indices c_e derived from the CRITIC-based assessment. Path continuity and flow conservation constraints were enforced in strict accordance with the mathematical formulation detailed in Section 3.3. Furthermore, path branching was explicitly prevented. Each enterprise node was restricted to a maximum of one incoming and one outgoing selected link. The main parameters used in the Evolutionary Solver are presented in Table 3.

Table 3. Evolutionary Solver parameters used in the case study

Parameter	Value
Population size	500
Mutation rate	0.2
Convergence	0.0001
Maximum time without improvement	120 s
Decision variables	Binary

The max average criticality model's minimum path-length constraint ($L_{\min}$ = 4) is intended to preclude the selection of trivially short routes consisting of only a few highly critical links that would be optimal with respect to the average criticality objective. By adopting a 4-link minimum, the model is forced to identify a route that covers an appropriately sized portion of the multi-tier supply chain. This constraint provides a way of defining the scope of the route. Each model was run 5 times with different random number seeds to account for stochastic variation. The model solver settings were identical for each run. Each of the 5 runs for each model resulted in the same optimal solution, so that solution was chosen as the final solution for each model. The minimum path-length constraint was included to prevent the model from selecting routes containing too few highly critical links.

4.4 Maximum Cumulative Criticality Model

In the first optimization scenario, cumulative criticality was defined as the objective function. Five links: L_{35}, L_{33}, L_{21}, L_{9}, L_{2}, were identified by the Evolutionary Solver. Based on these supplier-customer relationships, the continuous critical path is defined as:

$$\mathrm{S}_{19} \rightarrow \mathrm{S}_{18} \rightarrow \mathrm{S}_{16} \rightarrow \mathrm{S}_{9} \rightarrow \mathrm{S}_{3} \rightarrow \mathrm{S}_{1}$$

Criticality values corresponding with the selected links were calculated as 1.000, 0.878, 0.711, 0.473, and 0.213, respectively. Using the unrounded criticality indices, a total cumulative criticality of 3.274 was obtained by the model, corresponding to an average criticality score of 0.655.

Accumulation of criticality across several consecutive supplier–customer relationships is inherently favored by the cumulative model. Consequently, isolated high-criticality links are not simply selected; rather, the most critical feasible combination within a continuous path is identified.

4.5 Maximum Average Criticality Model

In the second optimization scenario, average link criticality across the selected route was maximized. Under the specified minimum length constraint of $L_{\min}$ = 4, four links were selected by the Evolutionary Solver: L_{35}, L_{25}, L_{14}, L_{5}.

The resulting critical path is expressed as:

$$\mathrm{S}_{19} \rightarrow \mathrm{S}_{18} \rightarrow \mathrm{S}_{11} \rightarrow \mathrm{S}_{6} \rightarrow \mathrm{S}_{1}$$

Corresponding criticality values for these constituent links were determined as 1.000, 0.775, 0.608, and 0.355, respectively. Using the unrounded criticality indices, a cumulative path criticality of 2.738 was obtained, yielding a maximum average criticality of 0.684. Compared with the first scenario, the shorter route obtained in the second scenario has a higher concentration of criticality among its links. In other words, the average criticality value is higher compared with the first scenario.

4.6 Comparison and Discussion of the Optimization Results

The primary parameters and empirical outcomes of both optimization scenarios are summarized in Table 4.

Table 4. Comparative summary of optimization scenarios and parameters

Parameter	Maximum Cumulative Criticality	Maximum Average Criticality
Objective function	Z_1	Z_2
Minimum number of links	-	4
Number of selected links	5	4
Cumulative criticality	3.274	2.738
Average criticality	0.655	0.684
Critical path	$\mathrm{S}_{19} \rightarrow \mathrm{S}_{18} \rightarrow \mathrm{S}_{16} \rightarrow \mathrm{S}_{9} \rightarrow \mathrm{S}_{3} \rightarrow \mathrm{S}_{1}$	$\mathrm{S}_{19} \rightarrow \mathrm{S}_{18} \rightarrow \mathrm{S}_{11} \rightarrow \mathrm{S}_{6} \rightarrow \mathrm{S}_{1}$

Note: "Z" denotes the objective function value, "S" denotes a supplier, and "-" indicates that no minimum path length constraint was imposed.

A longer path featuring higher cumulative criticality is mapped by the initial optimization formulation. Conversely, a shorter route characterized by a higher concentration of average link criticality is isolated under the second model. Consequently, cumulative criticality across a sequence of supplier-customer relationships is captured by the cumulative formulation, while a path characterized by a higher concentration of critical links is highlighted by the average formulation.

Crucially, selection of link L_{35} corresponding to relationship $\mathrm{S}_{19} \rightarrow \mathrm{S}_{18}$ with a maximum criticality value of 1 is selected by both optimization models. Downstream of this shared initial segment, divergence between the two paths is observed. Material flow progresses through intermediate supplier enterprises L_{16}, L_9, and L_3 under the cumulative model, whereas continuation through L_{11} and L_6 is dictated by the average model. Thus, the $S_{19} \rightarrow S_{18}$ relationship is identified as a common highly critical segment, as its systemic importance is validated independently of the chosen optimization objective.

Furthermore, the necessity of evaluating individual link criticality in conjunction with supply chain topology is demonstrated by these findings. Although a maximum individual criticality of $c_{32} = 1$ was calculated for link L_{32}, this edge is excluded from both optimal paths. High standalone criticality does not automatically guarantee inclusion within the most critical continuous route. Optimal path selection is governed simultaneously by individual link criticality metrics, supply chain topology, path continuity constraints, and the mathematical properties of the selected objective function.

Accordingly, valuable analytical support for managerial decision-making is provided by this dual-model framework. Inter-organizational relationships included across both optimal solutions can be prioritized for targeted risk mitigation and operational improvement. Simultaneously, alternative vulnerability patterns that might otherwise remain hidden under a single optimization perspective are revealed by the structural differences between the two identified paths.

5 Conclusions

An integrated methodology for critical path identification in multi-tier supply chains was developed by combining the CRITIC method with evolutionary optimization. The proposed approach first evaluates individual supplier–customer relationships using multiple performance criteria and then uses the resulting criticality indices to identify continuous critical paths within the supply chain network. In this way, the criticality of individual relationships is considered together with their position and connectivity within the network.

The methodology was applied to an automotive supply chain comprising 19 enterprises and 35 directed supplier–customer links. The 35 links were evaluated using 14 performance criteria, and the CRITIC method was used to determine objective criteria weights and calculate the criticality index for each relationship. The obtained indices varied considerably across the network, indicating that supplier–customer relationships do not contribute equally to supply chain criticality. At the same time, the results showed that the ranking of individual links cannot be used alone to identify the most critical continuous path. The position of a link within the network and the continuity of the selected route also affect the resulting path.

Two optimization models were subsequently applied. The maximum cumulative criticality model identified a five-link path with a cumulative criticality of 3.274 and an average criticality of 0.655. The maximum average criticality model, subject to the minimum path-length constraint, identified a four-link path with a cumulative criticality of 2.738 and an average criticality of 0.684. These results illustrate the different perspectives offered by the two formulations. The cumulative model favors the accumulation of criticality across a longer continuous route, whereas the average model identifies a route in which criticality is more concentrated among the selected links.

The comparison of the two solutions also provided an important structural finding. Both optimization models selected the same highly critical initial relationship, after which the identified paths diverged. This common segment can therefore be regarded as a particularly critical segment in the analyzed supply chain, as its inclusion does not depend on the selected path criticality formulation. In contrast, another link with the maximum individual criticality was not included in either optimal path. This finding further demonstrates that a high criticality index at the link level does not necessarily imply high importance at the path level. Critical path identification therefore requires the individual criticality of relationships to be considered together with supply chain topology, path continuity, and the selected optimization objective.

From a managerial perspective, the proposed approach can support the prioritization of supplier–customer relationships and continuous supply chain segments for further analysis and risk mitigation. Relationships appearing in both optimal paths may receive particular attention, while the comparison of cumulative and average criticality provides decision-makers with two complementary views of supply chain vulnerability. The identified critical segments can consequently be considered when planning monitoring activities, risk mitigation measures, and supplier development initiatives.

Several limitations should be considered when interpreting the results. The empirical analysis was conducted in a single automotive supply chain, and the evaluation of supplier–customer relationships was based on assessments provided by individual managers from the customer perspective. The resulting criticality indices therefore depend on the selected performance criteria and the available managerial assessments. In addition, the identified paths depend on the structure of the analyzed supply chain and the formulation of the optimization models. Future research could extend the approach to larger and dynamically changing supply chain networks, incorporate uncertainty arising from multiple decision-makers, consider time-dependent risk propagation, and investigate alternative evolutionary and metaheuristic optimization approaches in different industrial contexts.

Author Contributions

Conceptualization, D.T. and D.M.; methodology, D.M. and D.T.; validation, D.M.; formal analysis, D.T.; investigation, D.T.; resources, D.M.; data curation, D.T.; writing—original draft preparation, D.T.; writing—review and editing, D.M.; visualization, D.T.; supervision, D.M. All authors have read and agreed to the published version of the manuscript.

Data Availability

The data supporting our research results are included within the article.

Conflicts of Interest

The authors declare no conflicts of interest.

Declaration on the Use of Generative AI and AI-assisted Technologies

In the conduct of the research and the preparation of this paper, the authors did not use generative AI or AI-assisted technologies.

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